Computer Networks
Project

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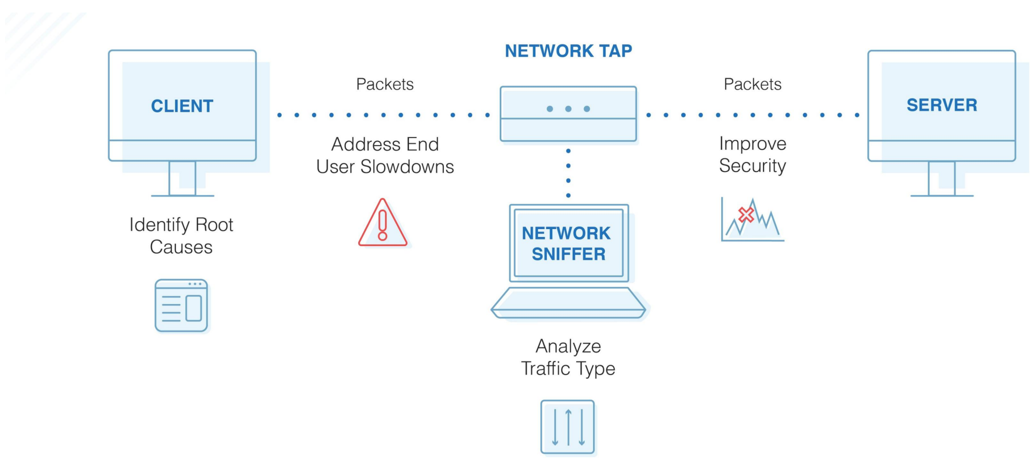
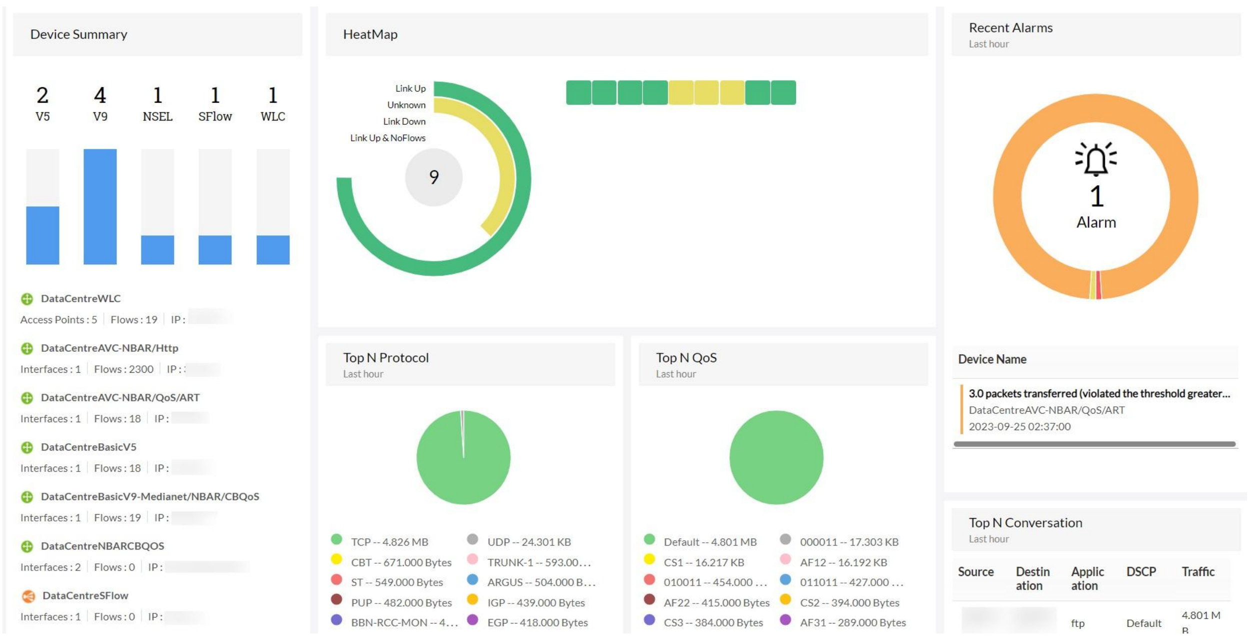
Title
Packet Sniffer and Network Traffic Analyzer using Python





Title Page

- Project title:
  - Packet sniffer and Network Traffic analyzer using python
- Course name:
  - Computer Networking
- Group member names:
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- Instructor name:
  - Miss Poma Panezai
- Department name:
  - FICT (Data Science)
- Submission date:
  - 13/JULY/2026

Introduction

- Packet sniffing:

Packet sniffing is a technique of monitoring every packet that crosses the network. The tools commonly used to do packet sniffing techniques are generally Wireshark and Netcut. Packet sniffing is usually done by hackers or malicious intruders to carry out prohibited actions such as stealing passwords and retrieving other important data. Then for the way the packet sniffing works is divided into three processes, namely collecting, conversion, analysis, and data theft.

- Packet sniffer:

It is a hardware and software tool that monitors network traffic. It is used to capture and analyze data packet which is sent between devices on Computer Network.

- Why Network Analysis Important:

Network analysis is very important because:

1. It improves Network security by detecting unauthorized network activity.
2. It ensures smooth communication between devices and applications.
3. It optimizes Bandwidth usage to ensure efficient use of Network.
4. It detects faults early, to reduce downtime and improve reliability.
5. It troubleshoots network problems such as packet loss and delays.
6. It monitors network performance by identifying slow or congested.

Objectives

- The main goals of project are:
- To capture network packets from network interface.
- To analyze packet information such as source IP, Destination IP, Protocol and Packet size.
- To identify network protocols (TCP, UDP, ICMP, HTTP, DNS etc.)
- To monitor network traffic to understand how data

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flows between devices.

- To display captured packet details in a simple format.
- To store captured packet data in file for later analysis.

Tools and Technologies

- We use the following Tools in our project:
  - Python
  - Wireshark
  - Github
- We used the following library in project:
  - Scapy
  - Matplotib
  - Panda
  - Datetime

Methodology

We implement and design Packet sniffing project in following steps:

Step No-1: Analysis step:

- We identified main objectives of project.
- We used python as programming language.
- We used Scapy, Matplotib, Panda and Datetime library to capture, analyze, generate graphs and timestamp the collected data.

Step No-2: Designing step:

- We designed the workflow of application.
- We divided project into modules:
  - Packet capture
  - Packet analysis
  - Packet Graphs
  - Packet Storage

Step No-3: Packet Capture:

- We capture incoming and outgoing network packets in real time using scapy library.

Step No-4: Packet Analysis:

- In this step we extracted information of packets.
  - Source Ip address.
  - Destination Ip address.
  - Protocol (TCP, UDP).
  - Port number if available.

Step No-5: Display result:

- The analyzed packet information is displayed on screen in readable format.
- This allows us to graphically watch the network traffic.

Step No-6: Stored Data packet:

- We stored capture packet information in CSV, TXT file and can be reviewed later.

Step No-7: Testing and Validation:

- We tested the application on network.
- We verified that the captured packet is correct.
- We checked that analysis and storage function works properly.

Implementation

Our project is divided into different modules/ methods. Each module/ method performs specific task.

- packet_sniffer.py module:

1. It captures packets from selected network interface.
2. It uses the scapy library.
3. It uses sniff() function to listen network traffic.
4. It passes captured packet to callback function for process.

- analyzer.py module:

➢ It analyzes the capturing packet to extract information.

1. It analyzes source IP address.
2. It analyzes destination IP address.
3. It analyzes protocol (TCP, UDP).
4. It analyzes packet size.
5. It analyzes source and destination ports.

- visualizer.py module:

1. It generate visual graphs and pie charts for the analysis.

Results and Screenshots

Figure 1: Protocol Distribution (Bar chart):

The bellow graph shows the distribution of captured packets based on transport layer protocols. From graph it is clear that TCP packets are the most common.

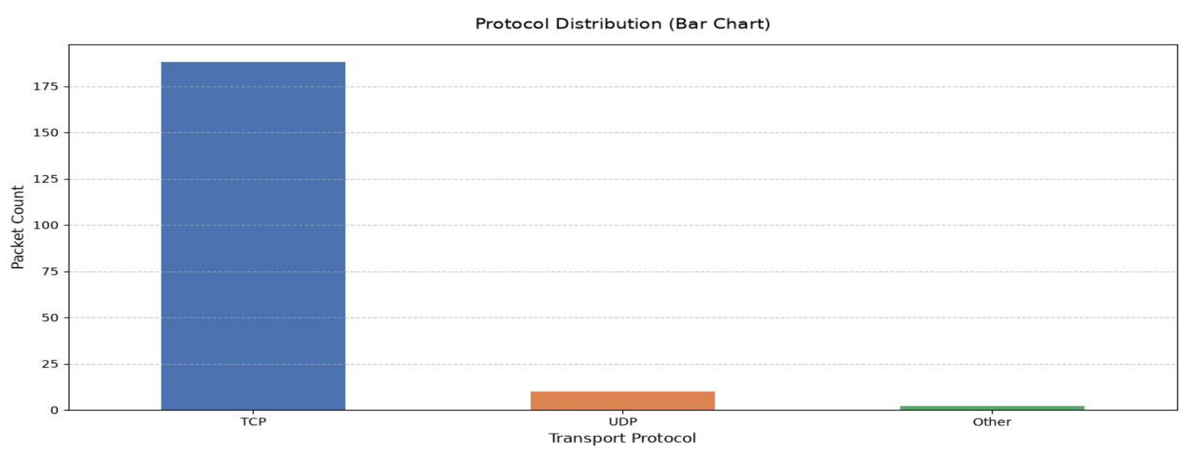

Figure 2: Protocol Distribution (Pie Chart):

The Pie chart shows the distribution of captured packet in percentage based on transport layer protocols. This chart shows that TCP approximately capture 94% of packets and about 5% of packets are captured by UDP.

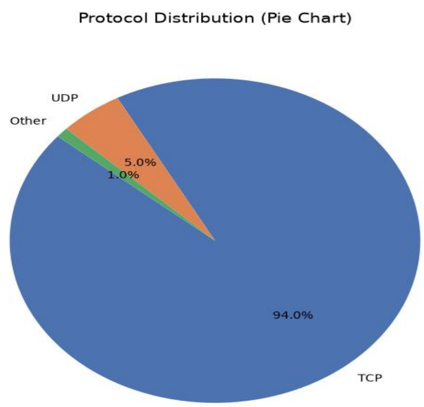

Figure 3: Top 5 Destination ports (Bar chart):

The bellow bar chart displays the top five destination ports identified during the Packet capture. The graph show that port 443 has the highest number of captured Packets, more than 100 packets.

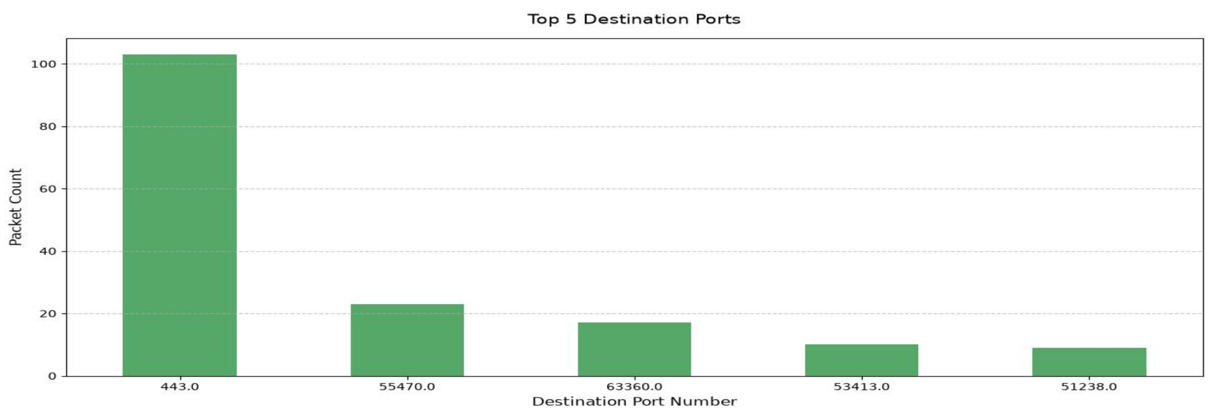

Figure 4: Top 5 source IP addresses (Bar chart):

The bar chart displays the top five source IP addresses observed during the packet capture. The graph indicates that the IP Address **192.168.100.224** generated the highest amount of network traffic, approximately 70 packets.

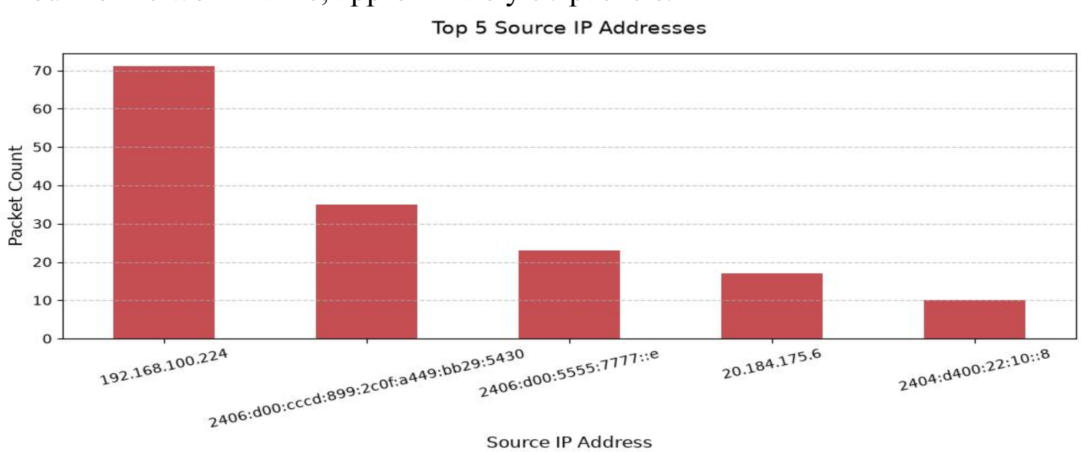

Figure 5: Network Traffic analyzer Output:

The graph below shows the final output of the packet sniffing.The analyzer processed 200 network packets captured from the network and generate summary of traffic.

```
=====================================================
      NETWORK TRAFFIC ANALYZER
=====================================================

[*] Total Packets Analyzed: 200

[*] Protocol Breakdown:
-------------------------
Protocol
TCP      188
UDP       10
Other      2

[*] Top 5 Most Active Source IPs:
-----------------------------------
Source_IP
192.168.100.224                              71
2406:d00:cccd:899:2c0f:a449:bb29:5430       35
2406:d00:5555:7777::e                        23
20.184.175.6                                 17
2404:d400:22:10::8                           10

[*] Top 5 Most Active Destination IPs:
---------------------------------------
Destination_IP
2406:d00:cccd:899:2c0f:a449:bb29:5430       49
20.184.175.6                                 43
192.168.100.224                              37
2406:d00:5555:7777::e                        11
13.69.116.107                                10

[*] Top 5 Most Used Destination Ports:
---------------------------------------
Destination_Port
443.0      103
55470.0     23
63360.0     17
53413.0     10
51238.0      9
```

Conclusio

After working on this project, we understand how computers communicate with each other. How data travels from one device to another device. This project also improved our practical knowledge of using python. This project was helpful to practically learn computer networking and network monitoring. This project provides strong foundation for future work in computer networking.

References

We used some books and websites for information are given below:

- Books we use:
  - Python Crash Course.
  - K.W computer Networking.
- Websites we used:
  - Scapy official Website.
  - Python official documentation.

Github Repository Link:

MianRasool/packet-sniffer-network-analyzer